

On rainbow connection

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Abstract

An edge-colored graph G is *rainbow connected* if any two vertices are connected by a path whose edges have distinct colors. The *rainbow connection number* of a connected graph G , denoted $rc(G)$, is the smallest number of colors that are needed in order to make G rainbow connected. In this paper we prove several non-trivial upper bounds for $rc(G)$, as well as determine sufficient conditions that guarantee $rc(G) = 2$. Among our results we prove that if G is a connected graph with n vertices and with minimum degree 3 then $rc(G) < 5n/6$, and if the minimum degree is δ then $rc(G) \leq \frac{\ln \delta}{\delta} n(1 + o_\delta(1))$. We also determine the threshold function for a random graph to have $rc(G) = 2$ and make several conjectures concerning the computational complexity of rainbow connection.

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1 Introduction

All graphs in this paper are finite, undirected and simple. We follow the notation and terminology of [2]. Connectivity is perhaps the most fundamental graph-theoretic property. There are many ways to strengthen the connectivity property, such as requiring hamiltonicity, k -connectivity, imposing bounds on the diameter, requiring the existence of edge-disjoint spanning trees, and so on.

A natural and interesting quantifiable way to strengthen the connectivity requirement was recently introduced by Chartrand et al. in [6]. An edge-colored graph G is *rainbow connected* if any two vertices are connected by a path whose edges have distinct colors. Clearly, if a graph is rainbow connected, then it is also connected. Conversely, any connected graph has a trivial edge coloring that makes it rainbow connected; just color each edge with a distinct color. Thus, one can properly define the *rainbow connection number* of a connected graph G , denoted $rc(G)$, as the smallest number of colors that are needed in order to make G rainbow connected. An easy observation is that if G has n vertices then $rc(G) \leq n - 1$, since one may color the edges of a given spanning tree with distinct colors, and color the remaining edges with one of the already used colors or, as we shall equivalently and conveniently assume throughout this paper, leave the remaining edges uncolored. Chartrand et al. computed the precise rainbow connection number of several graph classes including complete multipartite graphs [6]. We note also the trivial fact that $rc(G) = 1$ if and only if G is a clique, the (almost) trivial fact that $rc(G) = n - 1$ if and only if G is a tree, and the easy observation that a cycle with $k > 3$ vertices has rainbow connection number $\lceil k/2 \rceil$. Also notice that, clearly, $rc(G) \geq \text{diam}(G)$ where $\text{diam}(G)$ denotes the diameter of G .

Our goal in this paper is to study the extremal graph-theoretic behavior of rainbow connection. Motivated by the fact that there are graphs with minimum degree 2 and with $rc(G) = n - 3$ (just take two vertex-disjoint triangles and connect them by a path of length $n - 5$), and by the fact that cliques have $rc(G) = 1$, it is interesting to study the behavior of $rc(G)$ with respect to the minimum degree $\delta(G)$. Our main results relate the minimum degree of a graph with its rainbow connection number.

Is it true that minimum degree at least 3 guarantees $rc(G) \leq \alpha n$ where $\alpha < 1$ is independent of n ? This turns out to be correct, although certainly not trivial. Indeed, we prove:

Theorem 1.1 *If G is a connected graph with n vertices and $\delta(G) \geq 3$ then $rc(G) < 5n/6$.*

The constant $5/6$ appearing in the proof of Theorem 1.1 is not optimal, but we are unable to improve it significantly. In fact, it provably *cannot* be replaced with a constant smaller than $3/4$, since, as we shall see, there are 3-regular connected graphs with $rc(G) = \text{diam}(G) = (3n - 10)/4$. In fact, we conjecture:

Conjecture 1.2 *If G is a connected graph with n vertices and $\delta(G) \geq 3$ then $rc(G) < 3n/4$.*

We are able to assert Conjecture 1.2 in the case of 3-regular class-1 graphs (recall that a graph G is class-1 if its chromatic index $\chi'(G)$ is equal to its maximum degree). This is

a special case of the following theorem that improves the bound in Theorem 1.1 in many cases.

Theorem 1.3 *Suppose G is a connected graph with n vertices, and assume that there is a set of vertex-disjoint cycles that cover all but s vertices of G . Then $rc(G) < 3n/4 + s/4 - 1/2$. In particular:*

1. *If G has a 2-factor then $rc(G) < 3n/4$.*
2. *If G is k -regular and k is even then $rc(G) < 3n/4$.*
3. *If G is k -regular and $\chi'(G) = k$ then $rc(G) < 3n/4$.*

Not surprisingly, as the minimum degree increases, the rainbow connection number decreases. Specifically, we can prove the following upper bound.

Theorem 1.4 *Let G be a connected graph with n vertices and minimum degree δ . Then,*

$$rc(G) \leq \min\left\{ n \frac{\ln \delta}{\delta} (1 + o_\delta(1)) , n \frac{4 \ln \delta + 3}{\delta} \right\}.$$

Already for $\delta = 18$, Theorem 1.4 gives a better bound 0.81 for $rc(G)$ than the bound 0.833 from Theorem 1.1. We do not know how far Theorem 1.4 is from being tight, but in any case it cannot be improved below $\frac{3n}{\delta+1} - \frac{\delta+7}{\delta+1}$ as there are connected n -vertex graphs with minimum degree δ and this diameter.

Our next two results give non-trivial sufficient conditions for having $rc(G) = 2$. As noted earlier, having diameter 2 is a necessary requirement for having $rc(G) = 2$, although certainly not sufficient (e.g., consider a star). Clearly, if $\delta(G) \geq n/2$ then $diam(G) = 2$, but we do not know if this guarantees $rc(G) = 2$. The next theorem shows that by slightly increasing the minimum degree assumption, $rc(G) = 2$ follows.

Theorem 1.5 *Any non-complete graph with $\delta(G) \geq n/2 + \log n$ has $rc(G) = 2$.*

Another intriguing question is the *random graph* setting. Let $G = G(n, p)$ denote, as usual, the random graph with n vertices and edge probability p . In the extensive study of the properties of random graphs, many researchers observed that there are sharp *threshold functions* for various natural graph properties. For a graph property A and for a function $p = p(n)$, we say that $G(n, p)$ satisfies A *almost surely* if the probability that $G(n, p(n))$ satisfies A tends to 1 as n tends to infinity. We say that a function $f(n)$ is a *sharp threshold function* for the property A if there are two positive constants c and C so that $G(n, cf(n))$ almost surely does not satisfy A and $G(n, p)$ satisfies A almost surely for all $p \geq Cf(n)$. It is well known that all monotone graph properties have a sharp threshold (see [3] and [7]). Since having $rc(G) \leq 2$ is a monotone graph property (adding edges does not destroy this property), it has a sharp threshold function. The following theorem establishes it.

Theorem 1.6 *$p = \sqrt{\log n/n}$ is a sharp threshold function for the property $rc(G(n, p)) \leq 2$.*